

INDU 211: Introduction to Production & Manufacturing Systems

Final Exam Quantitative Problem Guide: Operations Research, Queuing, Quality Control & PERT

Department of Mechanical, Industrial & Aerospace Engineering · Concordia University

Executive Summary & Final Exam Alignment

In the Concordia INDU 211 curriculum (under Dr. Masoumeh Kazemi Zanjani), the **Final Examination (50% of Total Grade)** is comprehensive but heavily concentrated on the post-midterm topics taught in **Weeks 7 through 12**:

- **Week 7:** Deterministic Operations Research Models & Linear Programming (Chapter 14)
- **Week 8:** Probabilistic Operations Research Models & Queuing Theory (Chapter 15)
- **Weeks 9–10:** Quality Control, Control Charts ($\bar{X} - R$) & Process Capability (Chapter 8)
- **Week 11:** Work Design, Work Measurement & Human Factors (Chapters 6 & 11)
- **Week 12:** Project Management, Network Scheduling, CPM & PERT (Chapter 17)

From the official INDU 211 Problem Solutions series, **five video problems** directly cover the core quantitative problem types that appear on the Final Exam:

1. **Video 2: Minimization Problem** ([ytI70c0_cq8](#)) —

Chapter 14: Deterministic OR Models

. Graphical method for linear programming, equality constraints, 1D line-segment feasible regions, and corner-point evaluation.

2. **Video 6: Modeling Problem** ([Rwc_f6IzUQk](#)) —

Chapter 14: Deterministic OR Models

. Multi-product linear programming formulation, profit maximization, material capacity, machine hour equivalence, demand thresholds, and proportional production ratios.

3. **Video 4: Queuing Theory Problem** ([XT1EgQRcqmu](#)) —

Chapter 15: Probabilistic OR Models

. $M/M/1$ queuing systems, traffic intensity, expected waiting and system times, multi-customer congestion probabilities, and daily operating performance.

4. **Video 7: Quality Control Problem** ([1BcAZosLMb0](#)) —

Chapter 8: Quality Control

. $\bar{X} - R$ control charts, out-of-control point elimination, revised process mean, process standard deviation estimation ($\hat{\sigma} = \bar{R}/d_2$), and process capability ratio (C_p) evaluation.

5. **Video 3: PERT Chart** ([b2g1kZrEYtk](#)) —

Chapter 17: Project Management

. Activity-on-Node (AON) network diagrams, precedence constraints, path enumeration, critical path identification, and project completion duration.

Below is the exhaustive mathematical dissection, step-by-step solutions, and exam scoring frameworks for each problem.

MODULE 1: Linear Programming Minimization with Equality Constraints (Video 2)

Curriculum Context: Chapter 14 — Deterministic Operations Research Models

- **Video Reference:** [INDU 211 - Minimization Problem](#) (Duration: 19:44)
- **Exam Weight:** High-yield 15–20 mark question on Graphical Linear Programming

1.1 Mathematical Formulation

The linear programming problem requires minimizing total cost subject to an equality constraint, an inequality constraint, and non-negativity:

Minimize $Z = 4X_1 + X_2$ Subject to: $X_1 + 3X_2 = 9$ (Constraint 1: Equality) $X_1 + X_2 \leq 5$ (Constraint 2: Capacity) $X_1 \geq 0, X_2 \geq 0$ (Non-

1.2 Graphical Analysis & 1D Line-Segment Feasible Region

Step 1: Determine Boundary Intercepts

- **Constraint 1** ($X_1 + 3X_2 = 9$):
 - Set $X_1 = 0 \implies 3X_2 = 9 \implies X_2 = 3 \implies (0, 3)$
 - Set $X_2 = 0 \implies X_1 = 9 \implies (9, 0)$
 -

Crucial Realization

: Because this is a strict **equality** ($=$), feasible points must lie **strictly ON** this line, not in a half-plane area.

- **Constraint 2** ($X_1 + X_2 \leq 5$):
 - Set $X_1 = 0 \implies X_2 = 5 \implies (0, 5)$
 - Set $X_2 = 0 \implies X_1 = 5 \implies (5, 0)$
 - Feasible points lie on or below this line ($X_1 + X_2 \leq 5$).
- **Non-Negativity** ($X_1 \geq 0, X_2 \geq 0$): Restricts the solution space strictly to the First Quadrant.

Step 2: Determine Extremities (Corner Points) of the Feasible Region

Because of the equality constraint, the feasible region collapses from a 2D polygonal area into a **1-dimensional line segment** along $X_1 + 3X_2 = 9$:

- **Extreme Point A (Left Bound)**: At the X_2 -axis where $X_1 = 0$:

$$0 + 3X_2 = 9 \implies X_2 = 3 \implies \text{Point } A = (0, 3)$$

Check Constraint 2: $0 + 3 = 3 \leq 5$ (Satisfied).

- **Extreme Point B (Right Bound)**: At the intersection of the two constraint boundaries:

$$\begin{cases} X_1 + 3X_2 = 9 \\ X_1 + X_2 = 5 \end{cases} \implies X_1 = 5 - X_2$$

Substitute X_1 into Constraint 1:

$$\begin{aligned} (5 - X_2) + 3X_2 &= 9 \implies 5 + 2X_2 = 9 \implies 2X_2 = 4 \implies X_2 = 2 \\ X_1 &= 5 - 2 = 3 \implies \text{Point } B = (3, 2) \end{aligned}$$

Check Constraint 1: $3 + 3(2) = 9$ (Satisfied).

1.3 Objective Function Propagation & Optimal Solution

Graphical Iso-Cost Line Propagation:

Set $Z = 4X_1 + X_2 = 5$ (arbitrary level):

- Intercepts: $(0, 5)$ and $(1.25, 0)$. Slope $= -4$.
- For a **minimization** problem, propagate the iso-cost line from the upper right inward toward the origin $(0, 0)$.
- The **first point** of the feasible line segment that the iso-cost line touches (or the lowest value corner point) is the optimal minimum.

Corner Point Evaluation:

Corner Point	Coordinates (X_1, X_2)	Objective Function Value $Z = 4X_1 + X_2$	Evaluation
Point A	$(0, 3)$	$Z = 4(0) + 3 = 3$	GLOBAL MINIMUM (OPTIMAL)
Point B	$(3, 2)$	$Z = 4(3) + 2 = 14$	Sub-optimal

- **Optimal Solution:**

$$X_1^* = 0, \quad X_2^* = 3, \quad Z^* = 3$$

Part 3.2.3: Mathematical Characterization of All Feasible Solutions

The question asks to identify

all

feasible solutions of this linear program:
$$S = \left\{ (X_1, X_2) \in \mathbb{R}^2; \begin{aligned} &X_1 + 3X_2 = 9, \quad 0 \leq X_1 \leq 3, \quad 0 \leq X_2 \leq 3 \end{aligned} \right\}$$

- **Written Explanation:**
"The feasible region consists of all points lying along the line segment $X_1 + 3X_2 = 9$ bounded between the endpoints $(0, 3)$ and $(3, 2)$."

MODULE 2: Multi-Constraint Linear Programming Production Modeling (Video 6)

Curriculum Context: Chapter 14 — Deterministic Operations Research Models

- **Video Reference:** [INDU 211 - Modeling Problem](#) (Duration: 18:11)
- **Exam Weight:** High-yield 15–20 mark LP formulation question on Final Exam

2.1 Problem Description & Technical Data

A pharmaceutical manufacturing company produces three distinct prescription drugs: **Drug A, Drug B, and Drug C**. The production process requires two specialized active ingredients (Ingredient 1 and Ingredient 2), specialized machine hours, minimum market demand thresholds, and strict proportional batch blending ratios:

- **Ingredient 1 Availability:** Maximum **4,000 grams** available.
- **Ingredient 2 Availability:** Maximum **6,000 grams** available.
- **Raw Material Usage Table (grams per unit):**

Ingredient	Drug A (x_A)	Drug B (x_B)	Drug C (x_C)	Available Supply
Ingredient 1	20 g	33 g	50 g	$\leq 4,000$ g
Ingredient 2	50 g	20 g	60 g	$\leq 6,000$ g

- **Machine Hour Availability:**
 - One unit of Drug A requires **twice** the machine time of Drug B ($t_A = 2t_B$).
 - One unit of Drug A requires **three times** the machine time of Drug C ($t_A = 3t_C$).
 - Total machine time available in the factory is sufficient to produce **1,500 units of Drug A** if exclusively dedicated to Drug A.
- **Market Demand Requirements:**
 - Minimum sales demand for Drug A: at least **300 units**.
 - Minimum sales demand for Drug B: at least **400 units**.
 - Minimum sales demand for Drug C: at least **250 units**.
- **Relative Production Ratio:**
 - Production volumes must be maintained in the exact ratio:

$$x_A : x_B : x_C = 2 : 1 : 3$$

- **Profit Margins:**
 - Profit per unit of Drug A = **\$40**
 - Profit per unit of Drug B = **\$30**
 - Profit per unit of Drug C = **\$60**

2.2 Complete Mathematical Model Formulation

Step 1: Decision Variables

Let:

- x_A = Number of units of Drug A produced
- x_B = Number of units of Drug B produced
- x_C = Number of units of Drug C produced

Step 2: Objective Function

Maximize total manufacturing profit:

$$\text{Maximize } Z = 40x_A + 30x_B + 60x_C$$

Step 3: Raw Material Capacity Constraints

- **Ingredient 1:**

$$20x_A + 33x_B + 50x_C \leq 4000$$

- **Ingredient 2:**

$$50x_A + 20x_B + 60x_C \leq 6000$$

Step 4: Machine Hour Equivalence Constraint

- **Theoretical Derivation:** We are not provided raw hours (e.g., minutes per tablet), but rather **relative time equivalence**:
 - $t_A = 2t_B \implies$ 1 unit of B consumes 0.5 equivalent units of A capacity.
 - $t_A = 3t_C \implies$ 1 unit of C consumes $\frac{1}{3}$ equivalent units of A capacity.
 - Total capacity = 1,500 equivalent units of A.
- **Constraint Formulation:**

$$x_A + 0.5x_B + \frac{1}{3}x_C \leq 1500$$

(Alternative integer form multiplying by 6: $6x_A + 3x_B + 2x_C \leq 9000$)

Step 5: Minimum Market Demand Constraints

To satisfy pre-committed customer orders:

$$x_A \geq 300$$

$$x_B \geq 400$$

$$x_C \geq 250$$

Step 6: Proportional Production Ratio Constraints

The production volumes must satisfy $x_A : x_B : x_C = 2 : 1 : 3$:

$$\frac{x_A}{2} = \frac{x_B}{1} = \frac{x_C}{3}$$

This yields two independent linear equality equations:

$$1. \frac{x_A}{2} = x_B \implies x_A - 2x_B = 0$$

$$2. \frac{x_C}{3} = x_B \implies 3x_B - x_C = 0$$

(Or alternatively: $3x_A - 2x_C = 0$)

Step 7: Non-Negativity Constraints

Physical units cannot be negative:

$$x_A \geq 0, \quad x_B \geq 0, \quad x_C \geq 0$$

MODULE 3: Queuing Theory ($M/M/1$) System Evaluation (Video 4)

Curriculum Context: Chapter 15 — Probabilistic Operations Research Models

- **Video Reference:** [INDU 211 - Queuing Theory Problem](#) (Duration: 17:23)
- **Exam Weight:** High-yield 15-mark quantitative problem on Queuing Models

3.1 Operating Parameters & $M/M/1$ Assumptions

A takeout fast-food restaurant operates a single drive-through service window open daily from 11:00 AM to 11:00 PM:

- **Operating Horizon:** $T = 12$ hours/day (720 minutes/day).
- **Customer Arrival Process:** Follows a Poisson distribution with an average inter-arrival time of 10 minutes:

$$\text{Mean Arrival Rate } \lambda = \frac{1 \text{ customer}}{10 \text{ minutes}} = \mathbf{6 \text{ customers/hour}}$$

- **Customer Service Process:** Follows an exponential service distribution with an average service time of 4 minutes:

$$\text{Mean Service Rate } \mu = \frac{60 \text{ minutes/hour}}{4 \text{ minutes/customer}} = \mathbf{15 \text{ customers/hour}}$$

- **Model Classification:** Single-server Poisson arrivals and exponential service times with infinite queue capacity and FIFO discipline: $M/M/1$ **Queuing System**.

3.2 Step-by-Step Problem Solutions

Part 5.1: Daily Busy Time of the Service Window

- **Step 1: Calculate Traffic Intensity (Server Utilization Factor ρ):**

$$\rho = \frac{\lambda}{\mu} = \frac{6}{15} = 0.40 \quad (40)$$

(Because $\rho = 0.40 < 1.0$, the queuing system is stable and will not experience infinite queue blow-up).

- **Step 2: Calculate Total Daily Busy Hours:**

$$\text{Daily Busy Time} = \rho \times T = 0.40 \times 12 \text{ hours} = \mathbf{4.80 \text{ hours/day}}$$

(Expressed in minutes: $4.80 \times 60 = 288 \text{ minutes/day}$)

Part 5.2: Expected Waiting Time in Queue ($\bar{W}_q$)

- **Formula:**

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)}$$

- **Calculation:**

$$W_q = \frac{6}{15(15 - 6)} = \frac{6}{15 \times 9} = \frac{6}{135} = \frac{2}{45} \text{ hours} \approx 0.0444 \text{ hours}$$

- **Convert to Minutes:**

$$W_q = 0.0444 \times 60 \text{ minutes} = \frac{120}{45} = \mathbf{2.67} \text{ minutes}$$

- **Interpretation:** An arriving customer waits in line an average of **2 minutes and 40 seconds** before being served at the window.
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Part 5.3: Expected Total Time in the System ($\bar{W}$)

- **Formula:**

$$W = W_q + \frac{1}{\mu} = \frac{1}{\mu - \lambda}$$

- **Calculation:**

$$W = \frac{1}{15 - 6} = \frac{1}{9} \text{ hours} \approx 0.1111 \text{ hours}$$

- **Convert to Minutes:**

$$W = \frac{1}{9} \times 60 \text{ minutes} = \mathbf{6.67} \text{ minutes}$$

- **Interpretation:** A customer spends an average of **6 minutes and 40 seconds** total at the restaurant (waiting in line + ordering, preparing food, and paying).
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Part 5.4: Congestion Analysis — Daily Time with ≥ 2 Customers in the System

- **Step 1: Probability Formula for n Customers in an $M/M/1$ System:**

$$P_n = (1 - \rho)\rho^n$$

- **Step 2: Probability of 2 or More Customers in the System ($P(n \geq 2)$):**

$$P(n \geq 2) = 1 - P_0 - P_1 = 1 - (1 - \rho)\rho^0 - (1 - \rho)\rho^1 = 1 - (1 - \rho) - \rho(1 - \rho) = \rho - \rho + \rho^2 = \rho^2$$

- **Calculation:**

$$P(n \geq 2) = (0.40)^2 = \mathbf{0.16} \quad (16\%)$$

- **Step 3: Calculate Estimated Daily Congestion Hours:**

$$\text{Daily Congestion Time} = P(n \geq 2) \times T = 0.16 \times 12 \text{ hours} = \mathbf{1.92} \text{ hours/day}$$

$$(\text{Expressed in minutes: } 1.92 \times 60 = 115.2 \text{ minutes/day})$$

MODULE 4: Statistical Quality Control ($\bar{X} - R$ Charts) & Process Capability (Video 7)

Curriculum Context: Chapter 8 — Quality Control

- **Video Reference:** [INDU 211 - Quality Control Problem](#) (Duration: 18:10)
 - **Exam Weight:** High-yield 15–20 mark question on Statistical Process Control (SPC)
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4.1 Problem Description & Technical Data

Bestwood Manufacturing produces short wooden structural beams with a nominal design length of 24.0 cm. Quality engineers gathered 15 sample subgroups ($m = 15$), with each subgroup consisting of 4 randomly sampled beams ($n = 4$):

- **Initial Process Statistics:**

- Number of subgroups: $m = 15$
- Subgroup sample size: $n = 4$
- Initial grand mean (average of averages): $\bar{\bar{X}}_{\text{initial}} = 24.00 \text{ cm}$
- Initial average range: $\bar{R}_{\text{initial}} = 1.413 \text{ cm}$
- **Control Chart Inspection:**
 - Construction of standard 3-sigma $\bar{X}$ and R charts revealed that **Sample 4** ($\bar{X}_4 = 26.15 \text{ cm}$) and **Sample 10** ($\bar{X}_{10} = 22.75 \text{ cm}$) exceeded the Upper and Lower Control Limits (UCL / LCL).
 - Assignable causes were identified and eliminated. Standard industrial quality protocol requires **eliminating the out-of-control subgroups** and recalculating revised in-control process parameters with the remaining $m = 13$ subgroups.
- **Engineering Design Specifications:**
 - Nominal length: $24.0 \text{ cm} \pm 2.2 \text{ cm}$
 - Upper Specification Limit: $\text{USL} = 24.0 + 2.2 = 26.2 \text{ cm}$
 - Lower Specification Limit: $\text{LSL} = 24.0 - 2.2 = 21.8 \text{ cm}$

4.2 Step-by-Step Problem Solutions

Part 4.1: Revised In-Control Average Length ($\bar{\bar{X}}_{\text{revised}}$)

- **Rapid Calculation Technique:** Rather than recalculating from 52 raw individual measurements:

1. Calculate initial sum of all 15 subgroup means: $\sum_{i=1}^{15} \bar{X}_i$
 $i = m \times \bar{X}$

$$\{\text{initial}\} = 15 \times 24.00 = 360.00 \text{ cm}$$

2. Subtract the two out-of-control sample means: $\sum_{\{\text{revised}\}} \bar{X}_i = 360.00 - \bar{X}_4 - \bar{X}_{10}$

$$\{10\} = 360.00 - 26.15 - 22.75 = 311.10 \text{ cm}$$

3. Divide by the revised number of in-control subgroups ($m_{\text{new}} = 15 - 2 = 13$):

$$\bar{\bar{X}}_{\text{revised}} = \frac{311.10}{13} = 23.93 \text{ cm}$$

- **Engineering Conclusion:** When the process is operating in a state of statistical control, the true average beam length produced is 23.93 cm.

Part 4.2: Estimation of Process Standard Deviation ($\hat{\sigma}$)

- **Theoretical Relationship:** In statistical process control, the population standard deviation is unbiasedly estimated from the average sample range using the Hartley factor d_2 :

$$\hat{\sigma} = \frac{\bar{R}}{d_2}$$

- **Constant Lookup:** From the Standard Control Chart Constants table for subgroup size $n = 4$:

$$d_2 = 2.059$$

- **Calculation:**

$$\hat{\sigma} = \frac{1.413}{2.059} = 0.686 \text{ cm}$$

Part 4.3: Process Capability Ratio (C_p) & Industrial Interpretation

- **Formula for Process Capability Ratio:**

$$C_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}} = \frac{\text{Total Tolerance Spread}}{\text{Process Natural Spread}}$$

- **Substitute Numerical Values:**

$$C_p = \frac{26.2 - 21.8}{6 \times 0.686} = \frac{4.40}{4.116} = 1.07$$

- **Industrial Engineering Interpretation & Quality Benchmark:**

- **The 1.33 Standard:** In modern manufacturing engineering, an existing process requires a minimum $C_p \geq 1.33$ (a 4-sigma safety margin) to be certified as "capable," and $C_p \geq 1.67$ for critical safety components.
- **Assessment:** Because $C_p = 1.07 < 1.33$, the beam manufacturing process is **NOT CAPABLE** of consistently producing parts within specification limits.
- **Practical Consequence:** A significant percentage of wood beams will fall outside the $\pm 2.2 \text{ cm}$ tolerance band, leading to high scrap rates, rework costs, and customer rejection. Bestwood engineering must either reduce process variance (σ) or negotiate wider engineering tolerances.

MODULE 5: Project Management & Critical Path Method / PERT (Video 3)

Curriculum Context: Chapter 17 — Project Management

- **Video Reference:** [INDU 211 - PERT Chart](#) (Duration: 6:52)
- **Exam Weight:** High-yield 10–15 mark scheduling question on Final Exam

5.1 Project Network Data Table

Consider an engineering product proposal project with seven distinct activities (A through G):

Activity	Description	Immediate Predecessors	Duration (Weeks)
A	Assess Customer Needs	None	2
B	Write & Submit Proposal	A	1
C	Obtain Client Approval	B	1
D	Preliminary Engineering Design	C	2
E	Prototype Fabrication	C	5
F	System Testing & Validation	D, E	5
G	Final Client Delivery	F	1

5.2 Activity-on-Node (AON) Network & Path Enumeration

Step 1: Trace Sequential Network Topology

- Project starts at **A** (Duration 2).
- Moves to **B** (Duration 1).
- Moves to **C** (Duration 1).
- At **C**, the network branches into two parallel paths:
 - Upper Branch: Activity **D** (Duration 2 weeks).
 - Lower Branch: Activity **E** (Duration 5 weeks).
- Both paths must merge at **F**: Activity **F** cannot commence until **both** D and E are 100% complete.
- Finally, activity **G** concludes the project.

Step 2: Enumerate All Paths from Start to Finish

Path Index	Sequence of Activities	Week Durations	Total Duration
Path 1	A → B → C → D → F → G	2 + 1 + 1 + 2 + 5 + 1	12 Weeks
Path 2	A → B → C → E → F → G	2 + 1 + 1 + 5 + 5 + 1	15 Weeks

5.3 Critical Path & Slack Analysis

- **Critical Path Identification:** The **Critical Path** is defined as the sequence of dependent activities with the **longest total duration** through the network. It dictates the minimum possible time required to complete the overall project.

Critical Path = A → B → C → E → F → G

Minimum Project Completion Time = 15 Weeks

- **Engineering Rationale for the Branching Decision (D vs. E):** Why does the critical path pass through E instead of D? Because Activity F requires *both*

D and E to be complete before testing can begin. Even though Activity D completes in only 2 weeks, testing cannot start because Prototype Fabrication (E) requires 5 weeks.

- Therefore, Activity D has a **Total Slack** of:

$Slack_D = 15 - 12 = 3 \text{ Weeks}$

- Activity D can be delayed up to 3 weeks without impacting the 15-week project deadline.
- In contrast, any delay in Activities A, B, C, E, F, or G will directly delay the final delivery to the client.